PHYSICAL CHEMISTRY

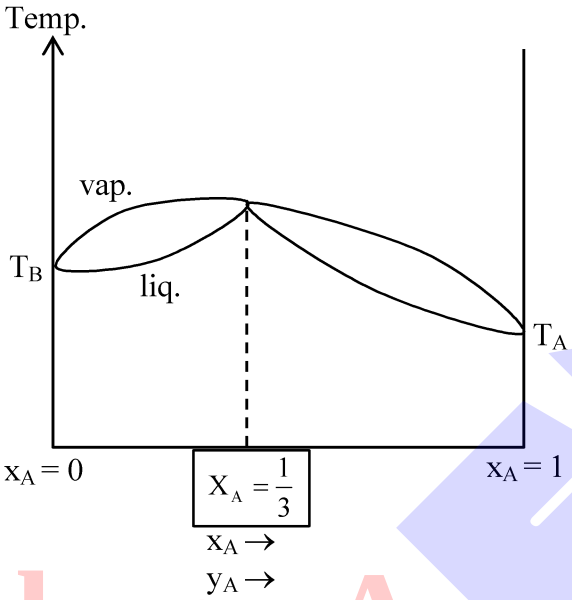
GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : LIQUID SOLUTION (Question Bank # 08)

SINGLE CORRECT

1. (C)

Sol. (1)



$p_A^o > p_B^o$

(A) $x_A = \dfrac{n_A}{n_A + n_B} = \dfrac{\frac{50}{100}}{\frac{50}{100} + \frac{50}{50}} = \dfrac{1}{3}$

$x_B = \dfrac{2}{3}$ $\quad \therefore$ A is correct statement

(C) This mix. would lie on left of azeotropic sol^n; hence upon distillation, the vapour would shift towards pure B

(D) This mix. would lie in right of azeotrop & hence upon distillation, the vapur would shift towards pure A

$\Rightarrow$ Ans. C

2. (A)

Sol.

$$\left.\begin{aligned} p_{O_2} &= \frac{20}{100}\times 10 = 3\cdot 3\times 10^7 \times \frac{n_{O_2}}{n_{H_2O}} \\ p_{N_2} &= \frac{79}{100}\times 10 = 6\cdot 51\times 10^7 \times \frac{n_{N_2}}{n_{H_2O}} \end{aligned}\right\} \text{ Dividing (1) \& (2)}$$

$$\Rightarrow \quad \frac{n_{O_2}}{n_{N_2}} \cong \frac{1}{2}$$

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3. (A)

Sol. $\Delta T_b = 2 = 0.76 \times \dfrac{\dfrac{2\cdot5}{M}}{0\cdot1} \times 1$

$\Rightarrow \quad M = \dfrac{0\cdot76\times25}{2} = 9.5 \text{ g/mol.}$

$\dfrac{\Delta p}{p^o} \simeq \dfrac{n_{solute}}{n_{solvent}} \quad \Rightarrow \quad \Delta p = 1 \text{ atm} \times \dfrac{2.5}{9.5/100/18}$

$\Rightarrow \quad \Delta p = 36 \text{ mm Hg.}$

$\Rightarrow \quad ps = 760 - 36 = 724 \text{ mm Hg}$

4. (D)

Sol. When half the solution has vaporited

$$p_s = \sqrt{p_A^\circ p_B^\circ} = \sqrt{121\times100} = 110 \text{ torr}$$

$\Rightarrow \quad 110 = p_A^\circ x_A + p_B^\circ (1 - x_A)$

$\Rightarrow \quad 110 = 100 \text{ torr} \times x_A + 121 \times (1 - x_A)$

$110 = 121 - 21x_A$

$$x_A = \frac{11}{21}$$

$\Rightarrow \quad n_A = x_A \times n_T = \dfrac{11}{21} \times 20 = \dfrac{220}{21}$

5. (B)

Sol. $\Delta T_f = k_f \times m \times i \Rightarrow \quad 0.2046 = 1.86 \times 0.1 \times i$

$\Rightarrow \quad i = 1.1 = 1 + (2-1)\alpha \Rightarrow \alpha = 0.1$

$\Rightarrow \quad K_a(HCOOH) = \dfrac{C\alpha^2}{1-\alpha} = \dfrac{0.1(0.01)}{0.9} = \dfrac{1}{9} \times 10^{-2}$

$\Rightarrow \quad k_h = \dfrac{k_w}{k_a} = \dfrac{10^{-14}}{\frac{1}{9}\times10^{-2}} = 9\times10^{-12} \text{ M}$

6. **(C)**

Sol. $4 = 1.86 \times \dfrac{0.75}{x} \Rightarrow \quad x = 0.34875 \text{ Kg}$

$\Rightarrow \quad$ ice separated $= (1 - 0.34875)$ kg

or

651.25 gm.

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7. **(A)**

Sol. $\pi = (C_2 - C_1)\,RT$

$\Rightarrow$ $3\text{atm} = (0.11 - 0.01) \times 0.08 \times T$

$\Rightarrow$ $T = 375\ K$

8. **(A)**

Sol. $\pi_{Total} = (\pi)_{urca} + (\pi)_{wall}$

new concentration of urca in sol$^n = \frac{1}{3}M$

new concentration of NaCl in sol$^n = \frac{0.6}{3}M$

$\Rightarrow$ $\pi_T = \left(\frac{1}{3} \times RT \times i\right)_{urca} + \left(\frac{0.3}{3} RT \times i\right)_{NaCl}$

$= RT\left(\frac{1}{3} + \frac{1.2}{3}\right) = 0.73RT$

9. **(A)**

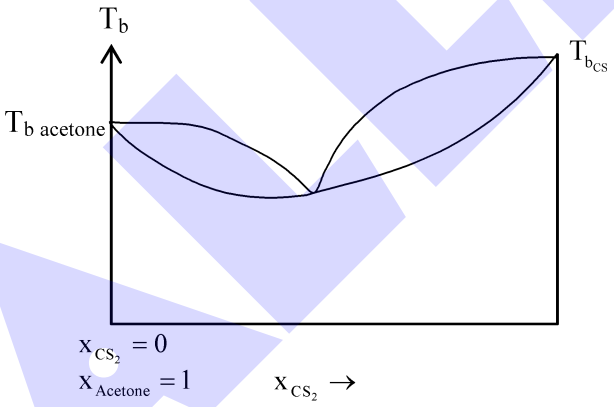
Sol. (A) most polar of all

MULTIPLE CORRECT

10. **(A,C,D)**

Sol. (A) can only be true for azeotropic sol^n.

(C)



When small amount of CS_2 is added, the boiling point decreases.

(D) Will from min. boiling azeotrop.

11. **(B,C,D)**

Sol. (B) Crenatin will take place when blood cells are placed in a more conc. sol^n than blood or hypertonic wrt blood

(D) Lysis will take place when blood cells are placed in a less concentrated sol^n than blood or lypotonic wrt. blood.

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12. (C)

Sol.

$$\Rightarrow \quad \begin{matrix} p_A^\circ > p_B^\circ \\ \frac{p_A^\circ}{p_B^\circ} > 1 \end{matrix} \quad \Bigg| \quad \begin{matrix} Y_A = \frac{p_A^\circ x_A}{p_s} \quad \text{——— (1)} \\ Y_B = \frac{p_B^\circ x_B}{p_s} \quad \text{——— (2)} \end{matrix}$$

Dividing (1) & (2);

$$\frac{y_A}{y_B} = \frac{x_A}{x_B}\left(\frac{p_A^\circ}{p_B^\circ}\right) \quad \Rightarrow \quad \frac{y_A}{y_B} > \frac{x_A}{x_B}$$

13. (A,B,D)

Sol. α of x in water = 1 ; $2x \rightleftharpoons x_2$

(for dimerisation) $\Rightarrow y = \frac{1}{2}$

α of x in water = 1 ; $x \longrightarrow y + z$

(for dissociation) $y = 2$

(A) $\Delta T_f = k_f \times \text{molality} \times i = 1.86 \times 1 \times 1 = 1.86$

$\Rightarrow T_f = -1.86\ ^\circ C$

(B) $\Delta T_b = k_b \times \text{molality} \times i = 0.52 \times 1 \times 2 = 1.04$

$\Rightarrow T_b = 101.04^\circ C$

(C) $7.44 = 1.86 \times \frac{1}{x} \times \frac{1}{2} \quad \Rightarrow \quad x = \frac{1}{8} = 0.125$ kg.

$\Rightarrow$ water separated as ice = (1 – 0.125) kg

= 0.875 kg

(D) $2.08 = 0.54 \times \frac{1}{y} \times 2 \Rightarrow \quad y = 0.5$ kg

$\Rightarrow$ water evaporated = (1 – 0.5) Kg.

= 0.5 kg.

14. (A)

Sol. $p^\circ \alpha (w_1 + w_2) = 0.24$ gm (increase in mass of absorbent)

$(p^\circ - p_s) \alpha w_2 = 0.02$ gm (decrease in mass of pure solvants)

$$\Rightarrow \quad \frac{p^\circ - p_s}{p^\circ} = \frac{0.02}{0.24} = \frac{1}{2} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

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$$\Rightarrow \quad \frac{n_{Glucose}}{n_{H_2O}} = \frac{1}{11} \quad \Rightarrow \quad \frac{wt_{Glucose}}{wt_{H_2O}} \times \frac{18}{180} = \frac{1}{11}$$

$$\Rightarrow \quad \frac{wt_{Glucose}}{wt_{H_2O}} = \frac{10}{11}$$

$$\Rightarrow \quad \text{mass \% of Glusose} = \frac{10}{21} \times 100 = \frac{1000}{21}\%$$

15. (A)

Sol. $n_{solute} \times i = 1 \times (1 + (4-1)(0.8)) = 3.4$

decrease in mass of sol^n = 0.18 g × p_s

increase in mass of absorbunt × $p_0 = w_1$

$$\Rightarrow \quad \frac{p_s}{p_0} = \frac{0.18}{w_1} = \frac{n_{solvent}}{n_{solvent} + n_{solute} x\, i}$$

$$\Rightarrow \quad \frac{0.18}{w_1} = \frac{17}{17 + 3.4} \quad \Rightarrow \quad w_1 = 0.216 \text{ g}$$

16. (B)

Sol. mass of solution would increase as the air would de-saturate & return solvent to liquid state.

17. (A)

Sol. $\Delta H_{mix} = 0$ for ideal sol^{n}s

PARAGRAPH

18. (C,D) **19. (B)**

Sol. $\Delta S_{mix} = -(n_A R \ln x_A + n_B R \ln x_B)$

$$= -R\left(1\ln\left(\frac{1}{3}\right) + 2\ln\left(\frac{2}{3}\right)\right)$$

$$= 3.8 \text{ cal/K}$$

$\Rightarrow \Delta G_{mix} = -T\Delta S_{mix} = (-300 \times 3.8)$ cal

$= -1140$ cal

Section II

(1) (200.00)

$$M_{avg} = \frac{M_{theo}}{i}$$

$$\Rightarrow \Delta T_f = K_f \times \text{molality} \times i = K_f \times \frac{n_{solute}}{wt_{solvent}(kg)} \times i$$

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$$\Rightarrow \Delta T_f = K_f \times \frac{1}{\frac{M_{theo}}{0.1}} \times i \Rightarrow 0.093 = \frac{1.86}{0.1} \times \frac{i}{M_{theo}}$$

$$\Rightarrow M_{avg} = \frac{1.86}{0.1 \times 0.093} = 200 \text{ g/mol}$$

(2) (2.00)

$$p_1 = K_H \left(\frac{\frac{4 \times 1}{RT}}{1L} \right) = 1 \quad \text{——— (1)}$$

$$p_2 = -K_H \left(\frac{\frac{4 \times V}{RT}}{0.5\,\ell} \right) = 4 \quad \text{——— (2)}$$

} Dividing (1) & (2)

$$\Rightarrow V = 2 \text{ ml}$$

(3) (4.00)

if (molality × i) finally is more than initial then resultant solution will have higher boiling point than the initial sol^n.

(i)

$$BaCl_{2(aq.)} + Na_2SO_{4(aq.)}(\text{added}) \longrightarrow BaSO_4 \downarrow + 2NaCl_{(aq.)}$$

a 2a

o

initially; molality × i = a × 3 = 3a

finally ; molality × i = 2a × 2 = 4a

(ii)

$$K_4[Fe(CN)_6]_{(aq.)} + 2CuSO_{4(aq.)} \longrightarrow Cu_2[Fe(CN)_6] \downarrow + 2K_2SO_4$$

a 2a

0

initially; molality × i = a × 5 = 5a

finally ; molality × i = 2a × 3 = 6a

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(iii)

$$Al_2(SO_4)_{3(aq.)} + 3BaCl_{2(aq.)} \longrightarrow 3BaSO_4 \downarrow + 2AlCl_{3(aq.)}$$

a — 2a

0

initially; molality $\times$ i = a $\times$ 5 = 5a

finally ; molality $\times$ i = 2a $\times$ 3 = 6a

(iv)

$$AgNO_{3(aq.)} + 2KCN_{(aq.)} \longrightarrow K[Ag(CN)_2]_{(aq.)} + KNO_{3(aq.)}$$

a — a — a

0

initially; molality $\times$ i = a $\times$ 2 = 2a

finally ; molality $\times$ i = a $\times$ 2 + a $\times$ 2 = 4a

$\therefore$ all (i), (ii), (iii), (iv) will qualify

(4) (4.00)

$$\frac{p_{Nitrobenjene}}{p_{H_2O}} = \frac{n_{C_6H_5NO_2}}{n_{H_2O}} = \frac{wt_{C_6H_5NO_2}}{wt_{H_2O}} \times \frac{18}{123}$$

$$\Rightarrow \quad \frac{wt_{C_6H_5NO_2}}{wt_{H_2O}} = \frac{3.6}{98.4} \times \frac{123}{18} = \frac{1}{4} = \frac{1}{x}$$

$$\Rightarrow \quad x = 4$$

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